

Separated intervals with reciprocal sum one

Abstract

For every sufficiently large k , we construct k nonadjacent integer intervals, each of length at least two, whose reciprocal sum is 1. The proof uses elementary prime estimates, the marriage theorem, and a short roots-of-unity argument.

1 Introduction

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Call a finite family of integer intervals in $\{2, 3, \dots\}$ a *configuration* if each interval has length at least two and successive intervals have at least one unused integer between them. Write $w(F)$ for its reciprocal sum and $\nu(F)$ for its number of intervals. We prove the following form of Erdős Problem 289 [2].

Theorem 1. *Every sufficiently large integer k admits a configuration F with $\nu(F) = k$ and $w(F) = 1$. All intervals have length two, three or four; those longer than two belong to a fixed finite set independent of k .*

Remark 1. *The second author's earlier solution [4] gives a stronger conclusion at the price of a longer argument: intervals of length two and three only, all inside $[1, 20k]$.*

Put $D_n = \text{lcm}(1, \dots, n)$ and $G_n = D_n^{-1}\mathbb{Z}/\mathbb{Z} \subset \mathbb{Q}/\mathbb{Z}$. Thus $1/a \in G_n$ precisely when every prime power dividing a is at most n . The group changes only at prime powers: if $q = p^\alpha$, then G_q/G_{q-1} has order p . We use this organization, also present in [4, 6], to realize whole sets $(\beta + G_n) \times I_n$ of attainable *residues and counts*: every residue in $\beta + G_n$ occurs at every integer count in I_n . All weights will be positive and below 2. Once the fixed rational translate β lies in G_n , residue zero therefore gives weight exactly 1.

2 Pairs supplying the successive quotients

Lemma 1. *For every sufficiently large prime power $q = p^\alpha$, there are $\gg q/\log q$ pairs $\{qm, qm + \varepsilon\}$, with $\varepsilon = \pm 1$, distinct $1 \leq m < q$, $p \nmid m$, and every prime power dividing $qm + \varepsilon$ smaller than q . Each pair lies in $[a - 1, a + 1]$, with distinct centers*

$$a \equiv 0 \pmod{8}, \quad cq^2/\log q \leq a \leq q^2$$

for an absolute $c > 0$. Each center occurs at at most three stages q .

Proof. For a sufficiently large fixed D , Chebyshev's estimates $\pi(x) \asymp x/\log x$ [3] supply $\gg q/\log q$ primes $b \in (q/D, q/16)$, excluding p . Set $d = 8$ for odd p and $d = 1$ for $p = 2$. For $r = db$, let $t \in [1, q - 1]$ be its inverse modulo q . Then

$$m_+ = \frac{rt - 1}{q}, \quad m_- = \frac{r(q - t) + 1}{q}, \quad m_+ + m_- = r.$$

Both coefficients lie in $(0, r)$, and at least one is coprime to p . Choose it. Its companion is $dbT = qm + \varepsilon$, with $T = t$ or $q - t$.

Only $O(1)$ carrier primes b can fail the required prime-power condition. Primes other than 2, b contribute only through $T < q$. A power of b at least q requires $T = jb$, $1 \leq j < D$, and hence $djb^2 \equiv \varepsilon \pmod{q}$. Each of these finitely many congruences has at most four roots: if a solution exists, division by it reduces the question to $z^2 \equiv 1 \pmod{p^\alpha}$, and one factors $(z - 1)(z + 1)$. For $d = 8$, a bad power of two forces T to be a multiple of the least power of two at least $q/8$. There are at most eight such $T < q$, each determining at most one $b < q$ per sign through $8bT \equiv \varepsilon \pmod{q}$. For $p = 2$, both b, T are odd. Discard these exceptions.

Each m comes from at most four carriers: a carrier divides $qm - 1$ or $qm + 1$, and each of these numbers, being below q^2 , has at most two distinct prime factors greater than q/D when $q > D^3$. After removing repetitions, retain the larger half of the coefficients. There remain $\gg q/\log q$, all $\gg q/\log q$. Use $a = qm + \varepsilon$ for odd p , and $a = qm$ for $p = 2$. These are distinct multiples of 8 with the stated bounds. Finally q is the largest prime-power divisor of qm ; thus a center can occur only at the largest prime-power stages of $a - 1, a, a + 1$. $\square$

Lemma 2. *For sufficiently large n , the subset sums of any $N \geq n^{3/4}$ distinct units modulo n cover $\mathbb{Z}/n\mathbb{Z}$.*

Proof. For a unit set A and $\zeta = e^{2\pi i/n}$, the number $R(b)$ of subsets summing to b satisfies

$$nR(b) = \sum_{t=0}^{n-1} \zeta^{-bt} \prod_{a \in A} (1 + \zeta^{ta}).$$

For $0 < t < n$, put $g = \gcd(t, n)$. The residues ta are nonzero multiples of g , each with at most g preimages. The two open end intervals of length $N/4$ contain at most $N/(2g)$ such multiples altogether. Hence at least $N/2$ members satisfy $\|ta/n\| \geq N/(4n)$, where $\|x\|$ is distance to the nearest integer. Using $|\cos \pi x| \leq e^{-2\|x\|^2}$, we obtain

$$\left| \prod_{a \in A} (1 + \zeta^{ta}) \right| \leq 2^N e^{-N^3/(16n^2)}.$$

The $t = 0$ term is 2^N , so $nR(b) \geq 2^N(1 - (n-1)e^{-n^{1/4}/16}) > 0$ for large n . $\square$

Every pair in Lemma 1 has weight in G_q . Modulo G_{q-1} , its weight equals u/q , where $u = m^{-1} \pmod{q}$: the companion lies in G_{q-1} , and

$$\frac{u}{q} - \frac{1}{qm} = \frac{um - 1}{qm}$$

has denominator dividing m , hence dividing D_{q-1} . Consequently any $s(q) = \lceil q^{3/4} \rceil$ of these pairs have subset sums covering G_q/G_{q-1} , by Lemma 2.

3 A finite starting family

Let B be a sufficiently large power of two. All conditions on its size below will be uniform in the later choices. Write $D_B = 2^e M$, M odd, and choose $N \geq e$ with $2^{N-3} \geq M$. Put $K = M2^N$. Reserve the following $t = 2(N-2)$ pairs:

$$[2^i + 1, 2^i + 2], \quad [M2^i + 1, M2^i + 2], \quad 3 \leq i \leq N. \quad (1)$$

Their possible left extensions are $[h, h+2]$, with distinct starts $h \equiv 0 \pmod{8}$. Thus all are separated, and their baseline weight σ is less than $\frac{1}{2}(1+1/M)$. An extension adds $1/h$. Multiplying by K , the two chains offer the increments $M2^j$ and 2^j , $0 \leq j \leq N-3$. They represent every integer $0 \leq z < K/4$: write $z = Mu + v$, where $u < 2^{N-2}$ and $0 \leq v < M \leq 2^{N-3}$, and use the binary digits of u, v .

Five single intervals provide the remaining coarse adjustment:

h	0	15	28	43	55
J_h	$[5, 6]$	$[4, 6]$	$[2, 3]$	$[2, 4]$	$[2, 5]$
$w(J_h)$		$11/30 + h/60$			

The successive gaps in $\{0, 15, 28, 43, 55, 60\}/60$ are at most $1/4$. Given $x \in G_B$, represented in $[0, 1)$, choose the greatest $h/60 \leq x$. The remainder belongs to $[0, 1/4) \cap K^{-1}\mathbb{Z}$, since $60 \mid D_B \mid K$. The chains supply it. Each resulting configuration therefore has

$$w = c_0 + x, \quad \nu = t + 1, \quad c_0 = 11/30 + \sigma < 9/10.$$

The small interval ends at most at 6, while every possible chain extension starts at least at 8, so separation is preserved.

Now take $2s(B)$ pairs from Lemma 1 at stage B , avoiding all possible seed supports. There are $\gg B/\log B$ candidates. Only $O(\log \log B)$ can conflict with the two chains: in the stage band, whose endpoint ratio is $O(\log B)$, each geometric chain has that many terms, uniformly in M, N . Each term forbids $O(1)$ centers. The small intervals add $O(1)$ exclusions. Thus enough pairs remain, and their total weight is $O(\log B/B^{5/4}) < 1/20$. Their weights lie in G_B , so adding any fixed r -element subset merely permutes the residues supplied by the seed. For $0 \leq r \leq 2s(B)$, this gives a finite family realizing

$$(\beta + G_B) \times [u_B, u_B + 2s(B)], \quad \beta = c_0 \pmod{1}, \quad u_B = t + 1. \quad (2)$$

All its weights are below $19/10 + 1/20 = 39/20$. Its supports and the rational translate β are now fixed.

4 Enlarging the residue group and the count range

We use one elementary observation. Suppose a family realizes $(\beta + G) \times [u, v]$, and a new pool is separated internally and from every old configuration. Its pair weights lie in $G' \supset G$; s pairs have subset sums covering G'/G , and there are m further pairs. If $v - u \geq s$, the unions realize

$$(\beta + G') \times [u + s, v + m]. \quad (3)$$

Indeed, for a target count $k \in [u + s, v + m]$, set $r = \max\{0, k - v\}$. Then $0 \leq r \leq m$ and $u + s \leq k - r \leq v$. Choose any r further pairs A . For a target residue $z \in \beta + G'$, choose a subset C of the first s pairs with $z - \beta - w(A) - w(C) \in G$. The old family supplies residue $z - w(A) - w(C)$ and count $k - r - |C| \in [u, v]$; adjoining A, C proves (3).

At each prime-power stage $q > B$, discard the centers from Lemma 1 conflicting with the initial family. Its two chains exclude $O(\log \log q)$ centers in the stage band; its remaining supports exclude $O(B^{3/4})$. These bounds are uniform in their positions and the chain lengths. Thus every stage still has at least $9s(q)$ centers when B is sufficiently large, since

$$B^{3/4} + \log \log q + s(q) = o(q/\log q) \quad \text{uniformly for } q > B.$$

For any finite cutoff $X > B$, Hall’s theorem assigns $3s(q)$ distinct centers to each stage $B < q \leq X$, without sharing centers. Indeed, a center belongs to at most three stages, so for any set S of stages,

$$|N(S)| \geq \frac{1}{3} \sum_{q \in S} \deg(q) \geq \sum_{q \in S} 3s(q).$$

Apply the marriage theorem to $3s(q)$ copies of each stage. Distinct centers on the $8\mathbb{Z}$ grid give separated pairs. Use $s(q)$ pairs for the quotient cover and $2s(q)$ as the further pairs in (3).

Starting from (2), the count interval is updated by

$$[u, v] \mapsto [u + s(q), v + 2s(q)].$$

These updates are always permitted. The next prime power after x is at most $2x$, because a power of two lies in that range; its s -value is at most $2s(x)$. The initial width $2s(B)$ therefore suffices for the first stage. If the old width W is at least $s(q)$, the new width is $W + s(q) \geq 2s(q)$, enough for the next stage. Successive count intervals overlap and their right endpoints tend to infinity. Every tail of the sequence consequently covers all sufficiently large integers. The intervals depend only on the quotas, although the chosen pools may depend on X .

Each stage- q pool has total weight $O(\log q/q^{5/4})$. Choose B so large that the sum of these bounds over all integers $q > B$ is below $1/20$, and all preceding uniform requirements hold. This choice precedes the initial construction: none of the bounds depends on its later chain lengths or positions. Every resulting configuration has weight below 2.

Finally choose $X_0 > B$ with $\beta \in G_{X_0}$, possible because $\bigcup_n G_n = \mathbb{Q}/\mathbb{Z}$. For every sufficiently large k , some cutoff $X \geq X_0$ has k in its count interval. The construction at that cutoff realizes G_X at count k . Choose residue zero: its weight is a positive integer below 2, hence 1. All intervals longer than two belong to the fixed initial family. This proves Theorem 1.

5 Formal verification

Theorem 1 has been verified in Lean 4 with Mathlib, following the proof above lemma by lemma. The terminal declaration `Erdos289.CLT.main_theorem`, in the directory `Erdos289CLT` of the repository [5], proves a proposition stating Theorem 1 verbatim; its axiom report lists only `propext`, `Classical.choice` and `Quot.sound`, and no result from the literature is assumed. The same repository contains the earlier formalization [4] of the stronger statement with intervals of length two or three inside $[1, 20k]$.

AI declaration

The proof in Sections 2–4 is a simplification, obtained with the help of GPT 6, of the arguments in [4, 6]. The Lean formalization was written with Claude and GLM models from a human-written skeleton of definitions and lemma statements.

References

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